DOCKHUB · DESIGN DECISION RECORD

DockHub recommendations accepted

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DOCUMENT CONTROL

Document ID	DKH-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dockhub		
Source	docs/decisions/0002-recommendations-accepted.md @ 5167e03		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Items left open on 2026-09-25 decided by Amish as recommended (DKH-DEC-001)
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all review recommendations, what changed in the repo and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted by Amish on 2026-09-25 for every item that carried a recommendation. Items with no recommendation (Table 2) were decided by Amish on 2026-10-02, as later recommended: "i approve your recommendations for all 555 open decisions." (DKH-DEC-001).

Context

DKH-DDR-001 and the review note (`docs/REVIEW.md`) listed the TRL 2 review items and the TRL 3 engineering proposals. Items with a recommendation had been adopted for TRL 3 work but stayed open for Amish's review. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item that carried a recommendation is therefore decided as recommended. Where the recommendation named one option among several, that option is the decision. Items with no recommendation are not decided by this record.

Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item (DKH-DDR-001 number)	Decision	What changed in the repo
1	Budget, R16 (1)	Option (a): build the four-bay cabinet and fit bays 1 and 2 for the first prototype. `budget_usd` stays at $1,200 and covers that fit; SwapCell packs are excluded	Already applied at TRL 3 (`project.yaml` , DKH-REQ-001 R16, `bom/bom.csv` , `cad/src/model.py` , DKH-DWG-001). Status wording updated in DKH-DDR-001 v0.2, DKH-REQ-001 v0.4 and `bom/bom-notes.md` . Budget unchanged: $1,200 before and after
2	Solar target, R6 (2)	Keep the 400 W canopy as shade and supplement; R6 relaxed from 20 % to 10 %	Already applied in DKH-REQ-001 R6; wording updated in DKH-REQ-001 v0.4 and DKH-PRC-001 v0.4. R6 is still not met (9.3 %)
3	Bay layout (3)	Four bays in one row at hand height	No change; already in DKH-PRC-001 and `model.py`
4	Charging architecture (4)	One certified charger and one CAN channel per bay	No change; already in DKH-PRC-001 and BOM lines 5 and 6
5	Fire separation (5)	Separate steel compartment per bay with a rear plenum venting through the roof	No change; already in DKH-PRC-001 and `model.py`
6	Doors during a fire alarm (6)	Bay doors stay locked	No change; already in DKH-PRC-001
7	Solar connection (7)	MPPT controller feeding bay 1, no hybrid inverter	No change to the design; DKH-CAL-001 v0.2 now cites the MPPT control rule as decided
8	Access (8)	Offline-first NFC access with an optional app	No change; already in DKH-PRC-001
9	Pack health gate (9)	Hold packs below 70 % state of health	No change; already in DKH-PRC-001
10	SwapCell interface (12)	Build to SwapCell interface v0.3	No change; already cited in DKH-REQ-001 R2 and DKH-PRC-001
11	TRL 3 engineering proposals (13)	Release packs at 80 % state of charge or more; chargers with a power factor of 0.9 or better; plinth flush with the body (1,000 x 500 mm); 15 min rider patience and the demand profile of DKH-CAL-001	No change to numbers; DKH-CAL-001 v0.2 and `sizing.py` now cite the release threshold as decided
12	Two-bay first prototype that cannot run the duty case (review note, TRL 3 session)	Accepted with item 1: the prototype demonstrates the station with one resident pack; the duty case needs the four-bay fit ($1,423)	Recorded here and in `docs/REVIEW.md`

No design change follows from these decisions: each was already built into the TRL 3 model, BOM, drawing and calculations. The GA drawing DKH-DWG-001 stays at Rev P1 (regenerated, geometry and notes unchanged), and DKH-CAL-001 results are unchanged.

The TRL 2 review made no recommendation to reword the `project.yaml` pitch or problem, so they are unchanged.

Items left open on 2026-09-25

Table 2. Items with no recommendation on 2026-09-25, decided by Amish on 2026-10-02 (DKH-DEC-001).

Item	Options	Decision, 2026-10-02
Pack ownership model (DKH-DDR-001 item 10)	Station-owned pool, rider-owned one-for-one exchange, or fleet packs; to follow co-design	A station-owned pool of packs exchanged one for one; to be confirmed in co-design
First pilot site and partner (DKH-DDR-001 item 11)	A delivery-worker group in a high-income city, or a moto-taxi hub in East Africa	A delivery-worker group in a high-income city: New York City first, Los Deliveristas Unidos (Workers Justice Project) as the first candidate partner to approach
How to close R6 (9.3 % against 10 %)	Accept about 9 %, route flat packs to bay 1 in daylight and accept occasional waits, feed more bays from the sun, or drop the target	R6 restated as about 9 %, normal routing kept; flattest-pack solar switching to be tried as a TRL 4 controller change
How to close R9 (charging pauses above about 29 to 39 °C ambient)	Accept daytime pauses in hot climates, add active cooling, narrow the design ambient, or raise the charge temperature limit with SwapCell	Keep the 45 °C charge limit and accept daytime pauses; R9 restated as full-rate charging up to about 30 °C ambient, with the cold-soak limit near -1 °C stated; a first pilot site with summer highs near that; active cooling only for a hot-climate site

Consequences

- Requirement status is unchanged: R6 and R9 not met; R8, R15 and R16 at risk; R7 and R14 not verifiable at TRL 3; nine met.
- Cross-repo actions (recorded in `docs/REVIEW.md`, not changed here): ask SwapCell for the pack wake time on the coded INTERLOCK loop, which sets the 2 s return check; note for SwapCell that multi-bay hosts need a CAN channel per bay because packs default to node 0.
- TRL 4 work (bench bay, propagation test, anchor and charger selection with datasheets, purchasing) stays on hold by Amish's instruction. `trl` and `trl_target` stay at 3.